When Is a Second Visual Encoder Worth Calling? A Budget-Aware Study of Visual Place Recognition

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***Abstract*—A second visual encoder can correct retrieval errors, damage correct matches, or spend computation on queries both models miss. We compare predicting correction minus damage with predicting fast-model error, using five first-encoder similarity statistics and frozen global descriptors. Models and validation thresholds are fixed before evaluating 650 Nordland holdout queries and 200 Gardens Point night queries. With EigenPlaces as the second encoder, a nominal 25% validation budget transfers to 78.92% invocation on the holdout and 15.50% externally. Holdout Recall@1 falls from CosPlace's 14.15% to 10.31%; external Recall@1 stays at 34.50%. With the original, weaker DINOv2 CLS model, benefit routing largely avoids harmful holdout calls, whereas error prediction damages accuracy. Across 107100 real online query records, invocation alone does not determine total latency. These results support a limited empirical conclusion: routing targets must account for model complementarity, but a small benefit predictor does not guarantee accuracy gain or a transferable budget. We report failures, frame-tolerance sensitivity and data ambiguity rather than universal superiority.**

***Index Terms*—visual place recognition, conditional computation, retrieval, evaluation**

I. Introduction

Visual place recognition (VPR) retrieves previously observed places from an image database. Its candidates can support relocalization or loop closure, but retrieval alone does not establish a geometrically verified pose [1]. Appearance change motivates using a second model when a fast descriptor struggles. The benefit depends on how the models' errors overlap: a difficult query may defeat both models, and replacing a correct fast prediction can introduce an error.

We ask a restricted operational question: before computing a second-model query descriptor, can five statistics from the first model's scores guide useful additional calls? We compare score-gap invocation, random selection, supervised error prediction, and supervised correction minus damage. The encoders remain frozen, but the selector is trained. A triggered query receives the second model's whole global ranking, without score fusion, temporal matching, local reranking or geometric verification.

Uncertainty estimation and predicting postprocessing effects have prior work. Our contribution is a constrained empirical comparison under the same available information, preserving a weaker second model and the single development-selected specialist replacement. We examine three failure modes: changed correction/damage opportunities, transfer of validation budgets, and discrepancy between invocation and real execution cost. Both final datasets use frozen models and thresholds; their results do not select another encoder or a revised gate.

II. Related Work

CosPlace [2] and EigenPlaces [3] provide specialist global descriptors. We also examine DINOv2's [4] normalized global CLS output, a general visual representation. Our fixed low-resolution condition is not a reproduction of the original papers' benchmark numbers.

Score-based uncertainty, including Statistical Uncertainty (SU) [5], estimates match reliability. Selective image matching [6] uses matching evidence to study when postprocessing helps. Sequence-matching receptiveness prediction [7] already distinguishes beneficial and harmful sequence-processing effects. We do not introduce correction/damage labels. Our gate observes a single query's first-encoder score statistics and selects a second global encoder before computing its query output. We inspected the available selective-matching and SMR code. A numerical SU comparator is omitted because its official implementation and complete numerical conventions have not been verified; this limits the comparison.

III. Method

A. Retrieval and Available Information

Let $D = \{d_j\}_{j=1}^{N}$ be an ordered gallery. Frozen encoder m produces $z_m(I) = f_m(I)/\|f_m(I)\|_2$, and exact cosine similarity is

$$s_m(q_i, d_j) = z_m(q_i)^\top z_m(d_j). \tag{1}$$

Candidates are ordered by decreasing score; ties retain manifest order. Ground-truth aligned frames are used only for training labels and evaluation. Frame indices never enter the online gate and are not treated as geographic coordinates.

For sorted A scores $s_{(1)} \geq \cdots \geq s_{(N)}$, the input is

$$u = \left[s_{(1)},\ s_{(1)} - s_{(2)},\ s_{(1)} - s_{(5)},\ \mu_{10},\ \sigma_{10}\right]^\top, \tag{2}$$

where μ_{10} and σ_{10} are the top-ten mean and population standard deviation. Retained features are standardized using training queries only. B outputs are unavailable at the decision point.

Offline: gallery descriptors; train labels → logistic heads; validation scores → budget thresholds.

Online: $q \to A \to s_A \to u \to g(u) > t$

False: return A's ranking.
True: $q \to B \to s_B \to$ return B's ranking.

No B query output or ground truth before the decision.

Fig. 1. The execution boundary. Gallery descriptors are prepared offline; query B encoding occurs only after a positive gate.

B. Supervised Correction and Damage Targets

Let $a_i, b_i \in \{0, 1\}$ denote top-one correctness on training queries. Construct

$$y_i^+ = \mathbf{1}[a_i = 0, b_i = 1], \qquad y_i^- = \mathbf{1}[a_i = 1, b_i = 0]. \quad (3)$$

Two logistic heads give estimates $p_+(u), p_-(u)$ and

$$g(u) = p_+(u) - p_-(u). \quad (4)$$

For exact conditional probabilities, the difference equals $\mathbb{E}[b - a \mid u]$. Our regularized estimates are not certified probabilities or a guarantee of benefit. Both heads use $C = 1$, LBFGS, no class weights, seed 0 and at most 2000 iterations; convergence is checked. The primary error control has error/correct heads with the same features and regularization. Complementary targets yield the same ordering as a single error head, also reported. Equal head counts do not create independent information.

C. Budget Thresholds and Ranking Selection

For validation budgets $\beta \in \{0.10, 0.25, 0.50\}$, a threshold allows at most $\lfloor \beta Q_{\text{val}} \rfloor$ calls. Under strict $g > t$, we use an adjacent selected/unselected score midpoint, with conservative ties. The benefit threshold is clipped to at least zero, so a budget need not be filled when estimated net gain is negative. This numerical boundary handling was corrected during development after detecting batch-versus- single-query roundoff effects; original runs remain available.

If triggered, B retrieves over the full gallery and replaces A's entire ranking; otherwise A is returned (Fig. 1). Thresholds are fixed on final queries; the final invocation fraction may differ from the validation budget. For Q evaluated queries,

$$R@1_{\text{route}} - R@1_A = (N_{\text{rescue}} - N_{\text{harm}})/Q. \quad (5)$$

IV. Experimental Protocol

A. Data and Leakage Control

Nordland-Part-2020 [8] provides 1750 summer and 1750 winter frames with matching numeric IDs; we evaluate this supplied frame alignment. All summer frames form the gallery. One-based winter ordinals 1–600 train the gate, 701–1000 select validation thresholds, and 1101–1750 form the 650-query holdout. Blocks 601–700 and 1001–1100 are unused buffers. Numeric filenames are sorted before assigning ordinals; they start at 251, so ordinals differ from raw filename numbers. Primary positives are references within five aligned frames; radii one and ten are sensitivity analyses using unchanged rankings and decisions. This is a fixed-map route-block holdout, not an independent unseen route.

The external test uses Gardens Point's 200 day-right references and 200 night-right queries. Exactly aligned frames are the primary positives. The tutorial's soft ground truth is an ambiguity/ignore mask; our radius-eight positives are explicitly a custom tolerance analysis. Full decoding, ID order, alignment and duplicate hashes were checked. Repeated images remain in their original splits; none are removed to improve results. In the holdout, 187 queries belong to 79 byte-identical groups; 542 unique hashes remain among 650 records. Gardens Point has one duplicated night-image pair. These dependencies limit interpretation of fine frame-index errors and independent-sample claims.

B. Models and Controls

A is CosPlace ResNet18/512. Initial B is DINOv2 ViT-S/14 with a normalized 384-dimensional final CLS output and no register tokens. After retaining its development failure, the single predeclared replacement is EigenPlaces ResNet50/512. All models use RGB, ImageNet normalization, and antialiased bilinear tensor resizing to 224×224. The full-image warp is a declared adaptation, not DINOv2's original classification crop. EigenPlaces retains official modules and pooling; a redundant constructor initializer download is bypassed before strict loading of the full checkpoint. Loaded tensors, source commits, hashes and row IDs are recorded. TF32 and cuDNN benchmarking are disabled.

Controls comprise A-only, B-only, random invocation, top1–top2 gap, the same-input A-error predictor and benefit prediction. Ablations remove the harm head or the top-ten standard deviation. Random seeds 0–4 are all reported as means and population standard deviations. Random costs are validation-matched only; final actual fractions may differ. A ground-truth oracle is diagnostic and never deployed. Recall@K, $K = 1, 5, 10$, counts queries with a positive in their first K candidates. We also report invocation, rescues, harms and signed Recall gain.

C. Actual Runtime

Measurements use an RTX 3060 Laptop GPU, PyTorch 2.6.0+cu124, batch one, twenty warmup queries and three complete query passes. CUDA is synchronized around each query. Times include warm-file decode, RGB, preprocessing, selected encoders, gate and exact retrieval. Model loading, gallery building and reference checks are excluded and accounted for separately, using warm-file construction and index setup diagnostics. Torch allocator peaks do not measure full-process memory; conditional policies retain both encoders. Cached accuracy simulation is not model latency. The unlocked GPU clock limits fine timing comparisons.

V. Results

A. Complementarity

Validation A reaches 71.33% R@1 and initial DINOv2 51.67%. B can rescue 29 errors and damage 88 successes

TABLE I
DEVELOPMENT VALIDATION R@1 (%). CONDITIONAL ROWS USE EXACTLY THE INDICATED EIGENPLACES INVOCATION FRACTION. NO FINAL-TEST NUMBERS.

Policy	10%	25%	50%
Benefit minus harm	73.00	75.67	82.00
A-error predictor	73.67	77.33	82.00
Top1–top2 gap	73.33	75.67	80.33
Rescue head only	73.33	77.33	83.00
Benefit without std	72.33	76.00	81.00
A-only (0% B)		71.33	
B-only (100% B)		83.33	

among 300 queries; the diagnostic oracle is 81.00%. The benefit policy calls B zero times at all budgets, retaining A accuracy. This is not an accuracy improvement. At 10%, error prediction rescues eight and damages four, reaching 72.67%. The original model pair is retained after examining the alternative.

EigenPlaces reaches 83.33% R@1, with 51 rescue and 15 harm opportunities; its oracle is 88.33%. Training A/B R@1 are 82.50%/86.67%, with 59 rescue and 34 harm opportunities among 600 queries. The different balance across blocks cautions against inferring transfer from the fitted estimates.

B. Correction and Real Execution

At 25%, benefit prediction rescues 18 queries and damages five, improving A by 4.33 points. The error predictor rescues 22 and damages four, improving by 6.00 points. Validation-matched random selection averages 73.87% R@1 with a 1.34-point population standard deviation. At 50%, benefit prediction avoids three harms relative to error prediction but loses three rescues; both reach 82.00%. Less damage alone does not demonstrate higher net accuracy (Table I).

Measured validation mean times for the EigenPlaces pair are 8.92 ms (A-only), 14.44 ms (B-only), 13.02 ms (25% benefit) and 13.38 ms (25% error). Each conditional policy performs 900 A forwards and 225 B forwards over three 300-query passes. Every decision and ranking matches the frozen reference, checked outside the timer. For the initial pair, zero-call benefit routing takes 10.19 ms versus A-only's 7.78 ms: fewer second-model calls do not eliminate selector cost. These sequential laptop measurements are not confidence intervals for small timing differences.

C. Frozen Final Accuracy

Table II shows substantial transfer failure. Holdout CosPlace R@1 is 14.15%, EigenPlaces 10.00%, and DINOv2 3.38%; the specialist that was stronger in development is weaker on this block. EigenPlaces offers only 17 rescue versus 44 harm opportunities, with 541 queries missed by both encoders. Its top-one oracle is 16.77%. Externally, CosPlace/EigenPlaces reach 34.50%/36.00%, with 31 rescue and 28 harm opportunities; the oracle is 50.00%. Oracle gaps measure available top-one complementarity, not learnable or deployable performance. All strategies and seeds are retained in the accompanying artifacts.

D. Budget Drift and Target Choice

With frozen EigenPlaces policies, the nominal 25% benefit setting calls B on 78.92% of holdout queries. Eleven rescues and 36 harms reduce R@1 to 10.31%, below A-only and marginally below error prediction's 10.46%. The corresponding error and gap invocation fractions are 82.00% and 65.08%, not equal test costs. We therefore compare the observed accuracy–cost points rather than asserting a matched test budget. The nominal 10% and 50% benefit settings transfer to 40.15% and 82.46% invocation (Table III).

In Gardens Point, the nominal 25% benefit setting instead calls 15.50% of queries. Six rescues and six harms leave R@1 at 34.50%; error prediction calls 11.00% and reaches 34.00%. Benefit at nominal 50% reaches 36.50% with 29.50% invocation, only one additional correct query over B-only. Removing standard deviation also reaches 36.50% at 23.00% invocation, whereas rescue-only at 25% reaches 34.50%. The feature and target ablations do not establish a uniform advantage. Five-seed nominal-random R@1 is $13.32 \pm 0.37\%$ on the holdout and $35.20 \pm 1.60\%$ externally; these are population SDs across seeds, not confidence intervals. Validation-matched random and nominal-random use the same EigenPlaces validation fraction, but neither enforces the learned gate's shifted test fraction.

E. Original Pair and Tolerance Sensitivity

The original DINOv2 pair demonstrates a narrower benefit of the target. The holdout has eight rescue and 78 harm opportunities. Benefit at nominal 25% makes two calls, neither changing correctness, and retains A's 14.15% R@1. Error prediction calls 82.00%, rescues six and damages 65, reducing R@1 to 5.08%. Externally, benefit makes 18 calls with five rescues and three harms, reaching 35.50%, versus error's 35.00%. Avoiding calls to a weak model is useful, but preserving A-only accuracy does not establish improvement over A or superiority at equal test cost.

Tolerance changes remain separately labeled, using identical frozen rankings and masks. At Nordland radii one/ten, A-only R@1 is 5.08%/19.69%, and EigenPlaces benefit-25 is 4.62%/13.08%. For Gardens Point's custom radius-eight positives, A/EigenPlaces/ benefit-25/error-25 reach 97.00%/91.00%/93.00%/94.00%. This differs markedly from exact-index results and is not the tutorial's soft-ignore evaluation. Fine frame-index performance should not be equated with geographic localization. Examples are the first query in manifest order of each correctness class, not the most attractive cases; their IDs, descriptor samples, scores and rankings are supplied.

F. Real Cost and Replay Checks

On the EigenPlaces holdout, A-only has the highest observed R@1 and the lowest measured mean, 7.56 ms; benefit-25 takes 13.62 ms and B-only 10.66 ms. Externally, benefit-25 takes 7.97 ms versus A-only's 5.84 ms and B-only's 8.92 ms, with the accuracy tradeoff above. Benefit-50 exceeds B-only by one correct query but takes 9.09 ms, providing no observed

TABLE II

SINGLE-ENCODER FINAL RECALL@1/5/10 (%). ALL USE THE SAME DECLARED 224-PIXEL ADAPTATION; THESE ARE NOT ORIGINAL-PAPER BENCHMARK REPRODUCTIONS.

	Nordland, $r = 5$			Gardens Point, $r = 0$		
Encoder	R@1	R@5	R@10	R@1	R@5	R@10
CosPlace R18	14.15	23.54	30.62	34.50	88.50	93.50
EigenPlaces R50	10.00	16.46	24.77	36.00	86.00	96.00
DINOv2 ViT-S/14 CLS	3.38	10.46	16.77	23.50	76.00	87.50

TABLE III

FROZEN FINAL EVALUATION WITH EIGENPLACES AS B. BUDGETS IN LABELS ARE VALIDATION SETTINGS; B COLUMNS SHOW ACTUAL FINAL INVOCATION. RANDOM ROWS GIVE FIVE-SEED MEANS AND POPULATION SD FOR R@1. TIMES ARE ACTUAL FULL-QUERY MEANS (MS), WITH UNLOCKED LAPTOP CLOCKS.

	Nordland holdout ($Q = 650$, $r = 5$)			Gardens Point ($Q = 200$, $r = 0$)		
Policy	R@1 (%)	B (%)	ms	R@1 (%)	B (%)	ms
A-only	14.15	0.00	7.56	34.50	0.00	5.84
EigenPlaces-only	10.00	100.00	10.66	36.00	100.00	8.92
Benefit (10%)	12.15	40.15	15.09	34.00	9.50	7.40
Benefit (25%)	10.31	78.92	13.62	34.50	15.50	7.97
Benefit (50%)	10.46	82.46	14.41	36.50	29.50	9.09
A-error (10%)	10.00	75.54	18.52	34.00	7.00	7.18
A-error (25%)	10.46	82.00	17.86	34.00	11.00	7.55
A-error (50%)	10.31	84.77	14.47	34.00	18.00	8.10
Gap (25%)	12.77	65.08	12.87	34.50	14.50	7.78
Rescue only (25%)	10.62	80.46	18.27	34.50	10.00	7.45
Benefit, no std (25%)	10.31	78.46	18.18	36.50	23.00	8.57
Random, nominal 25%	13.32 ± 0.37	23.60	12.91	35.20 ± 1.60	25.40	8.68
Random, validation-matched	13.32 ± 0.37	23.60	13.19	35.20 ± 1.60	25.40	8.63

mean-latency saving. Figure 2 presents actual forwards, not cached descriptor timings. Unlocked clocks and sequential policy execution prevent attributing every timing difference to the target or claiming stable fine-scale speedups. A reduced number of second-model calls alone is not an end-to-end cost guarantee.

Independent replay checks all 85000 final offline rows, 107100 real online rows and 510 strategy–radius evaluations. Descriptors are checked against model-state and manifest hashes, row IDs, unit norms and array hashes. Each timed pass verifies the causal decision and complete ranking against its frozen reference outside the timer. No final fitting or threshold calibration occurs. Gallery building and warm-file setup costs are recorded separately; both models remain resident for conditional execution, so invocation reduction is not a model-memory reduction.

VI. DISCUSSION

The benefit target separates two risks that A-error prediction conflates, but the estimate is useful only if those conditional relationships transfer. EigenPlaces rescue prevalence changes from 51/300 validation queries to 17/650 holdout queries, while harm opportunities remain material. A-only score distributions also change: the median top1–top2 gap falls from 0.01477 to 0.00500, and median top-ten standard deviation from 0.02477 to 0.00230. Similar top-one scores can coexist with poor frame-index correctness. These are descriptive associations, not causal proof that any single statistic explains the failure.

Repeated and visually similar station frames constrain the information available to image-only retrieval. Exact Gardens Point frames penalize nearby references that may depict similar places. We retain all queries and report the predeclared tolerances rather than widening the primary criterion after seeing results. Small fixed-route datasets, correlated frames, low-resolution adaptation, linear gates and unlocked laptop clocks limit generalization. The missing verified SU implementation limits the uncertainty comparison. We introduce neither correction/damage labels nor a universally better algorithm, and make no open-set, metric-pose, SLAM or robot-deployment claim.

VII. CONCLUSION

In this frozen-encoder study, predicting net correction can avoid damaging calls to a weak second model, but neither reliable accuracy gain nor validation-budget transfer follows. A development-selected specialist does not preserve its advantage on a held-out route block. Reporting actual invocation, signed corrections and real latency exposes these failures more clearly than invocation budgets or validation Recall alone. The results motivate testing complementarity and budget transfer before deploying conditional retrieval; they do not establish universal benefit.

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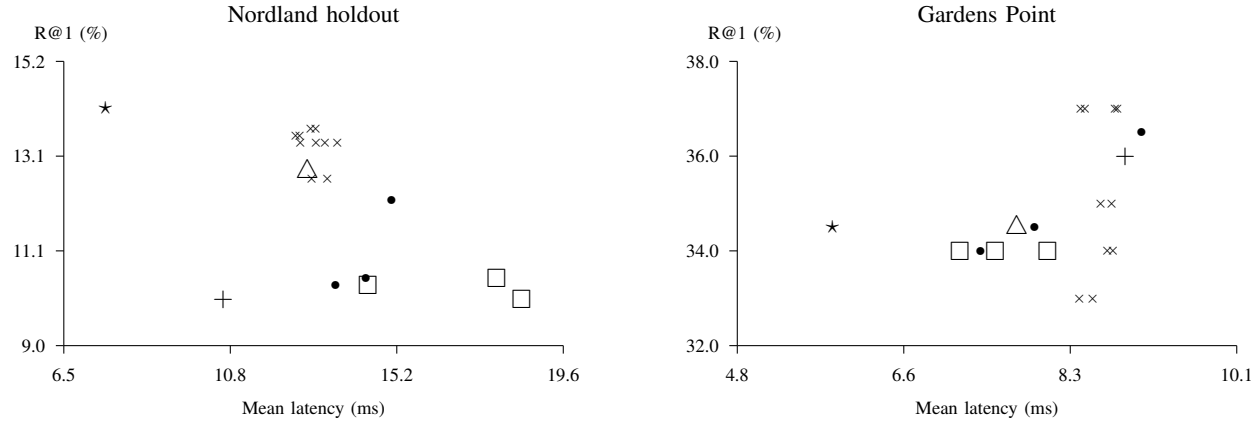


Fig. 2. Real execution of the EigenPlaces pair: star A-only, plus B-only, filled circles benefit and squares A-error at nominal 10/25/50%, triangle gap at 25%, and crosses all five seeds for each random control at 25%. Each point is measured; unmeasured policies are not interpolated. Actual calls appear in Table III.

Numerical values come from the logged executable experiments and replay checks. This working draft still requires human author review of interpretation, citations, authorship and the conference's disclosure and anonymity requirements.

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